When Does Unpaired Error Injection Help LLM Agents? Hypotheses from a Failed Transfer to Embodied Planning

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Abstract

Taxonomy-guided error injection yields +57pp system-level improvement for tool-use LLM agents (PALADIN; combining taxonomy-guided injection, recovery pairing, and inference-time retrieval), but does this transfer to embodied planning? We test TaxEI—a four-operator error taxonomy—on ALFWorld with Qwen3-8B, comparing clean training (Vanilla), taxonomy-tagged injection (TaxEI), and generic-labeled injection (Random) across three seeds. TaxEI yields no statistically significant improvement (-8.6pp, p=0.054). However, this comparison is confounded: PALADIN pairs errors with recovery demonstrations, while our TaxEI injects standalone errors. The negative result may reflect missing recovery pairing rather than task structure. Beyond this negative result, we observe a dose-dependent dual regime: at moderate injection (25%), performance degrades deterministically (-17.8pp, SD=1.2pp, 3 seeds), while at high injection (50%), extreme seed sensitivity emerges (SD≈12pp), with individual seeds ranging from 46.8% to 70.2%. We additionally observe a per-operator local/global split where local errors (+9.1pp) are learnable but global errors (-8.3pp) are harmful (exploratory, N=11–12). These patterns suggest that unpaired error injection may require discrete, observable, locally recoverable errors—conditions met in tool-use but not in embodied planning.

1 Introduction

Training LLM agents on failure trajectories is an increasingly popular paradigm. PALADIN (Vuddanti et al., 2025) demonstrates that taxonomy-guided error injection—where training data includes systematically corrupted trajectories tagged with error types—yields +57pp system-level improvement for tool-use agents (combining taxonomy-guided injection, recovery pairing, and inference-time retrieval). Related approaches show gains across tool-use (Song et al., 2024; Lan et al., 2025; Wang et al., 2024) and dialogue tasks (Ding, 2025). A natural question arises: does this transfer to embodied planning, where agents must execute multi-step physical manipulation tasks?

We test this on ALFWorld (Shridhar et al., 2021) with TaxEI, a four-operator error taxonomy for embodied planning, training Qwen3-8B agents under three conditions (Vanilla, TaxEI, Random) across three seeds. TaxEI yields no statistically significant improvement (-8.6pp, p=0.054) and directionally increases instability. Crucially, this comparison is confounded: PALADIN pairs each injected error with an explicit recovery demonstration, while our TaxEI injects standalone errors without paired recoveries. The negative result may therefore reflect missing recovery pairing rather than—or in addition to—differences in task structure.

Beyond reporting this negative result, we make two contributions:

1. **Negative transfer with competing hypotheses.** Unpaired error injection provides no statistically significant benefit on ALFWorld. Two equally plausible explanations exist: (a) embodied planning lacks the discrete, observable error structure that makes injection effective in tool-use, and (b) the absence of recovery pairing renders injected errors unlearnable. Disentangling these requires recovery-paired injection experiments.
2. **Dose-dependent dual regime.** At moderate injection (25%), performance degrades deterministically (-17.8pp vs. vanilla, SD=1.2pp, 3 seeds). At high injection (50%), extreme seed sensitivity emerges (SD≈12pp), with individual seeds ranging from 46.8% to 70.2%—the latter exceeding all vanilla seeds. The dose-response is non-monotonic: 50% mean (56.0%) exceeds 25% mean (47.5%).

We additionally observe a per-operator local/global split where local errors are learnable but global errors are harmful (exploratory, N=11–12), and a training-evaluation paradox at ≥25% injec-

tion. These exploratory patterns offer preliminary guidance for ALFWorld-like embodied tasks.

2 Method

2.1 Task Environment

ALFWorld (Shridhar et al., 2021) is a text-based embodied environment where agents execute multi-step household tasks (e.g., *heat a potato and place it in the fridge*). Each task requires a sequence of `think`/`act` steps in a ReAct-style loop (Yao et al., 2023). We use the standard 47-task unseen test split across six task types: `look_at_obj`, `pick_and_place`, `pick_clean`, `pick_cool`, `pick_heat`, and `pick_two_obj`.

2.2 Error Taxonomy

We define four error injection operators adapted for embodied planning, organized along two axes:

Local operators affect single steps without disrupting the overall plan structure:

- **C4 (observation_masking)**: Replaces environment observations with generic placeholders, removing perceptual grounding for one step.
- **P1 (step_order_swap)**: Swaps two adjacent action steps, creating a locally incorrect but recoverable sequence.

Global operators disrupt the trajectory's causal chain:

- **P2 (sub_goal_deletion)**: Removes a complete sub-goal (e.g., the "clean" step in a pick-clean-place task), breaking the plan's logical structure.
- **C1 (context_truncation)**: Truncates the context window, causing all subsequent decisions to rely on incomplete history.

All operators achieve 100% injection legality (valid action format) and 97.5% recovery quality (trajectories remain parseable). Error-recovery pairs are *not* explicitly paired in training data—errors are mixed as additional demonstrations.

2.3 Training Setup

We collect 501 expert trajectories from GPT-5.2 on ALFWorld training tasks, yielding 4,358 clean SFT demonstrations. Error injection produces 4,671 additional trajectories (round-robin across the four operators).

Three conditions:

- **Vanilla**: 4,358 clean demonstrations only.
- **TaxEI**: 4,358 clean + 4,671 taxonomy-tagged errors (error type labels in prompt) = 9,029 total.
- **Random**: 4,358 clean + 4,671 generic-labeled errors (no taxonomy) = 9,029 total.

All models fine-tune Qwen3-8B using LoRA (rank 16, α=32, lr 1×10^{-4}, 3 epochs, bf16). Each condition is trained with three seeds (42, 0, 123).

Dose-response ablation. To control for data volume effects, we fix total training size at 4,358 and vary the error injection ratio (10%, 25%, 50%). The 10% condition uses seed=42 only; 25% and 50% are evaluated across three seeds (42, 0, 123).

Recovery evaluation. We inject errors at inference time using round-robin operator assignment across 47 test tasks and measure whether error-trained models recover better than vanilla.

Method	N	s42	s0	s123	Mean±SD
Vanilla	4,358	61.7	68.1	66.0	65.3±3.3
TaxEI	9,029	61.7	46.8	61.7	56.7±8.6
Random	9,029	66.0	55.3	68.1	63.1±7.0

Table 1: Three-seed clean evaluation on ALFWorld (47 tasks, accuracy %). TaxEI vs. Vanilla: p=0.054; TaxEI vs. Random: p=0.170; Random vs. Vanilla: p>0.3 (pooled permutation, 10K). Note: TaxEI/Random use 9,029 samples (2.1× vanilla). The dose-response ablation (Table 2) controls for data volume at a fixed 4,358 total.

3 Experiments

We evaluate three training conditions on ALFWorld's 47 unseen test tasks (Shridhar et al., 2021): **Vanilla** (4,358 clean demonstrations), **TaxEI** (4,358 clean + 4,671 taxonomy-tagged error trajectories = 9,029), and **Random** (4,358 clean + 4,671 generic-labeled errors = 9,029). Training details are in §2.3. We report results across three seeds (42, 0, 123).

3.1 Main Results

Table 1 shows three-seed evaluation results. TaxEI degrades directionally (−8.6pp vs. Vanilla), but the difference is not statistically significant (permutation test, p=0.054, N=10,000). No TaxEI seed exceeds the Vanilla mean. Random injection shows minimal degradation (−2.2pp, p>0.3). The TaxEI–Random gap is driven by two of six task types (`pick_cool`: −19.0pp; `pick_two_obj`: −14.3pp).

Dose	s42	s0	s123	Mean±SD	Δ
TaxEI					
10%†	61.7	—	—	61.7	—
25%	48.9	46.8	46.8	47.5±1.2	−17.8
50%	51.1	70.2	46.8	56.0±12.4	−9.3
Random					
10%†	68.1	—	—	68.1	—
25%	55.3	61.7	51.1	56.0±5.3	−9.3
50%	57.4	46.8	70.2	58.1±11.7	−7.2

Table 2: Dose-response ablation (fixed 4,358 total). Δ: mean change vs. vanilla (65.3±3.3). †Single seed (seed=42); Δ omitted as baseline differs from the 3-seed mean used for other rows. At 25%, TaxEI degrades deterministically (SD=1.2pp). At 50%, variance explodes (SD≈12pp), with seeds spanning 46.8–70.2%.

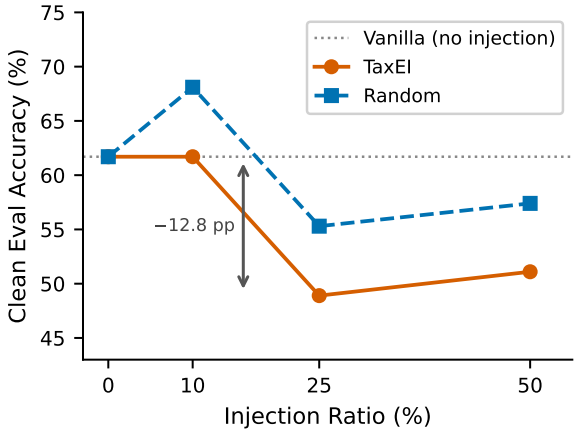


Figure 1: Dose-response (seed=42 only). 3-seed data in Table 2 reveals non-monotonic dose-response with regime bifurcation at 25%→50%.

At full training scale, standard deviations follow Vanilla (3.3pp) < Random (7.0pp) < TaxEI (8.6pp), though not statistically confirmed at n=3. The dose-response ablation (§3.2) reveals this variance is dose-dependent: deterministic at 25% (SD=1.2pp) but chaotic at 50% (SD≈12pp).

3.2 Dose-Response

To control for data volume, we fix total training size at 4,358 and vary the injection ratio (Table 2, Figure 1). Multi-seed results at 25% and 50% reveal a dual regime.

At 25%, TaxEI degrades deterministically: all three seeds converge to ∼47% (−17.8pp, SD=1.2pp). Random also degrades (−9.3pp, SD=5.3pp). At 50%, both methods exhibit extreme seed sensitivity (SD≈12pp), with seeds ranging from 46.8% to 70.2%—the latter exceeding all vanilla seeds. The dose-response is non-monotonic: 50% TaxEI mean (56.0%) exceeds 25% mean (47.5%). At 10% (single seed), TaxEI matches Vanilla exactly; Random achieves 68.1%, within vanilla seed range.

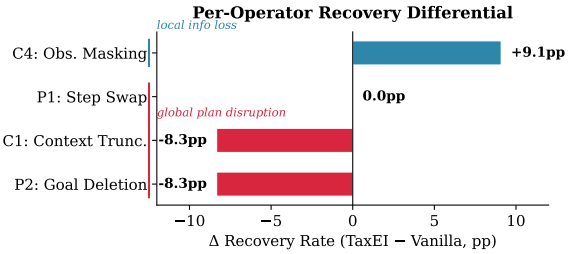


Figure 2: Per-operator recovery delta (TaxEI − Vanilla, seed=42). N=11–12; ±1 task ≈ ±8–9pp. Exploratory.

3.3 Recovery Evaluation

We test whether error-trained models recover from injected errors at inference time (round-robin operator injection, 47 tasks, seed=42 only, preliminary). TaxEI achieves 59.6% recovery (vs. Vanilla 61.7%, Random 63.8%)—no statistically significant improvement. The uniform 0% detection rate (seed=42) suggests taxonomy labels were not internalized during SFT—models learned trajectory patterns rather than error classification.

3.4 Per-Operator Local/Global Split

Figure 2 decomposes recovery by operator (seed=42, N=11–12). Local operators (C4 observation masking: 7/11 vs. 6/11, +9.1pp; P1 step swap: 7/12 vs. 7/12, 0.0pp) are neutral-to-beneficial; global operators (C1 context truncation: 8/12 vs. 9/12, −8.3pp; P2 sub-goal deletion: 6/12 vs. 7/12, −8.3pp) are harmful. Local errors leave the causal structure intact; global errors propagate downstream. These results are exploratory (N=11–12).

3.5 Training-Evaluation Paradox

At ≥25% injection, models can achieve *lower* training loss but *worse* evaluation performance. For seed=42, TaxEI-50% reaches final loss 0.259 (vs. Vanilla 0.288) yet evaluates at 51.1% (vs. 61.7%). This paradox does not hold at 10%, where training loss (0.308) remains above Vanilla. Two explanations are plausible: memorization of noise (Arpit et al., 2017; Zhou et al., 2023) and distribution shift at high injection ratios. The extreme seed variance at 50% suggests this paradox is seed-dependent: TaxEI-50% seed=0 achieves 70.2% despite similar training dynamics.

4 Discussion

Structural hypothesis. We hypothesize that the transfer failure stems from structural differences between tool-use and embodied planning. Table 3 (Appendix) identifies five dimensions where these domains diverge: error signal (discrete API codes vs. diffuse wrong-object errors), observability (explicit stack traces vs. silent failures), recovery scope (local retry vs. global replanning), feedback timing (immediate vs. delayed), and error locality (single-step vs. cascading). In tool-use, taxonomy labels encode a direct mapping from error type to recovery action. In embodied planning, errors propagate silently—picking the wrong object produces no error message, and recovery requires replanning from an unknown state. Reflexion (Shinn et al., 2023) achieves ∼97% on ALFWorld through inference-time failure reflection, suggesting that inference-time self-correction succeeds where training-time injection fails in embodied planning.

Recovery-pairing confound. PALADIN pairs each error with an explicit recovery demonstration via retrieval augmentation, contributing substantially to its +57pp system-level gain. Our TaxEI injects standalone errors without paired recoveries. We cannot distinguish between these two hypotheses with the current design: the negative result may stem from (a) missing recovery pairing, or (b) structural differences between tool-use and embodied planning, or both. Recovery-paired injection on ALFWorld is needed to disentangle these factors.

Training lottery at high injection. At 50% injection, individual seeds range from 46.8% to 70.2%—the latter exceeding all vanilla seeds. The 70.2% anomaly occurs in both TaxEI (seed=0) and Random (seed=123), suggesting a generic high-dose phenomenon independent of taxonomy labeling. This cross-condition coincidence may reflect a seed-initialization × injection interaction warranting further investigation. For practitioners, this implies that single-seed evaluation is unreliable when injection ratios exceed ∼25%: a minimum of three seeds is necessary to distinguish genuine improvements from seed-dependent outliers, and high injection ratios require particular caution.

Detection rate. The uniform 0% detection rate (§3.3) may reflect that SFT does not internalize taxonomy labels when they appear only as prompt prefixes—the training objective (next-token prediction) does not incentivize error classification, so models learn trajectory patterns rather than error type distinctions. This suggests a fundamental limitation of direct SFT mixing for error-awareness: without an explicit classification or contrastive objective, taxonomy labels serve as surface tokens that do not transfer to inference-time error recognition. Methods that use preference optimization (Song et al., 2024) may better incentivize error discrimination.

5 Related Work

Error injection for LLM agents. Several methods augment agent training with failure trajectories: PALADIN (Vuddanti et al., 2025) uses taxonomy-guided injection with retrieval-augmented recovery; ETO (Song et al., 2024) applies DPO on exploration failures; EEF (Lan et al., 2025) extracts positive signals from expert failures; Wang et al. (2024) and AgentHER (Ding, 2025) integrate contrastive or relabeled negatives. AgentDebug (Zhu et al., 2025) identifies five error dimensions including error cascading—a phenomenon our per-operator analysis independently corroborates. Their diagnostic taxonomy differs from our injection-oriented 4-operator design: AgentDebug classifies observed errors post-hoc, while our operators generate controlled perturbations. The overlap in cascading-error findings suggests this structural property is a robust moderator of error-injection effectiveness. FireAct (Chen et al., 2023) fine-tunes agents with diverse reasoning traces, showing that trajectory diversity improves robustness. Our work tests *cross-domain transfer* of the injection paradigm using direct SFT mixing.

Training dynamics and data mixing. McCoy et al. (2020) and Mosbach et al. (2021) document seed sensitivity in fine-tuning; D'Amour et al. (2022) frame this as underspecification. Our dose-dependent variance findings extend these observations to error-augmented agent training. Rolnick et al. (2017) show robustness to massive label noise in vision—contrasting with our sharp degradation at 25% injection in sequential decision-making. Ye et al. (2025) and Bengio et al. (2009) establish that data composition has non-linear effects on SFT performance.

ALFWorld methods. ReAct (Yao et al., 2023) establishes think/act prompting. Reflexion (Shinn

et al., 2023) achieves ∼97% via multi-trial inference-time self-reflection—notably, inference-time reflection succeeds where our training-time injection fails. CoEx (Kim and Hwang, 2025) reaches 93.3% with co-evolving world models; AgentPRM (Choudhury, 2025) achieves 91.0% with process reward models. LATS (Zhou et al., 2024) applies tree search over reasoning traces, achieving competitive ALFWorld results through inference-time planning. Our 65.3% vanilla baseline is consistent with single-trial SFT on an 8B model. No prior work has tested training-time error injection on ALFWorld.

6 Conclusion

We tested whether unpaired taxonomy-guided error injection—effective for tool-use agents—transfers to embodied planning. On ALFWorld, TaxEI provides no statistically significant improvement over clean training (-8.6pp, p=0.054). Two competing hypotheses explain this: embodied planning may lack the discrete, observable error structure required for injection to help, or the absence of recovery pairing (present in PALADIN) may render standalone errors unlearnable. We additionally identify a dose-dependent dual regime (3-seed confirmed at 25%/50%): deterministic degradation at moderate injection, chaotic seed-dependent outcomes at high injection. We also observe a per-operator local/global split (exploratory, N=11–12).

We hypothesize four properties that may moderate error injection effectiveness—all require independent validation, including on ALFWorld itself: (1) are errors discrete and observable? (2) can recovery be achieved locally? (3) does the task provide immediate diagnostic feedback (indicating error type or correction direction, not merely binary success/failure)? (4) are errors paired with explicit recovery demonstrations (untested in this work; inferred from PALADIN's design)?

Future work should first test recovery-paired injection on ALFWorld to disentangle the task-structure hypothesis from recovery-pairing effects, then replicate on a second environment (e.g., WebShop) and investigate the seed-initialization × injection interaction observed at high doses.

Limitations

We evaluate on a single environment (ALFWorld, 47 tasks) with one base model. Three seeds provide limited statistical power; the 10% dose-response uses a single seed (25%/50% confirmed across 3 seeds). TaxEI/Random use 2.1× more training data than Vanilla; the fixed-size ablation controls for this. Per-operator analysis (N=11–12) is exploratory. ALFWorld provides immediate binary feedback ("Nothing happens") but not diagnostic feedback indicating error type or correction direction, partially resembling tool-use error observability. Vision-based embodied tasks with continuous state spaces may exhibit different error injection dynamics. We compare only error injection variants; other augmentation strategies (e.g., paraphrasing, trajectory diversification) were not tested, leaving open whether the negative result generalizes to all off-distribution augmentation at this data scale.

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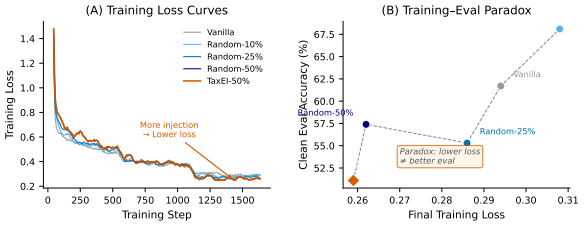


Figure 3: Left: training loss curves. Right: final loss vs. eval accuracy—lower loss corresponds to worse performance at ≥25% injection.

A Structural Comparison

Dimension	Tool-Use (Vuddanti et al., 2025)	Embodied
Error signal	Discrete (API error codes)	Diffuse (wrong object, wrong room)
Observability	Explicit (stack trace)	Silent (no error msg)
Recovery scope	Local (retry call)	Global (replan)
Feedback	Immediate	Delayed
Error locality	Single step	Cascading

Table 3: Structural comparison: tool-use vs. embodied planning. Error injection benefits from discrete, observable, locally recoverable errors.

B Training Dynamics

Figure 3 shows training loss curves across conditions. At ≥25% injection, models achieve lower final training loss than Vanilla (TaxEI-50%: 0.259 vs. Vanilla: 0.288) despite worse evaluation performance (§3.5). At 10% injection, training loss (0.308) remains above Vanilla.

C Per-Task Cross-Analysis

For seed=42, the ∼6.4pp aggregate gap between TaxEI and Random at each dose point corresponds to ≈3 tasks on a 47-task evaluation. This consistency does not hold across seeds: at 25% the 3-seed mean gap is 8.5pp, at 50% it is 2.1pp. Cross-dose analysis reveals this is a discretization coincidence: zero tasks appear as TaxEI-penalty in all three dose points (10%, 25%, 50%). The specific tasks that flip between conditions rotate across doses—9 different tasks contribute to the penalty, with no single task appearing at all three levels.

D Version Discrepancy

Our evaluation pipeline evolved from v3 to v4 during development. The v3 vanilla baseline (74.5%, seed=42) was higher due to a format mismatch.

The v3→v4 transition involved three changes: (1) prompt format alignment from ChatML to tool-format, (2) sample count increase from 4,176 to 4,358 due to improved trajectory collection, and (3) seed handling standardization. The 12.8pp drop (74.5%→61.7%) is attributable primarily to the format alignment. All three conditions (Vanilla, TaxEI, Random) were trained and evaluated under the same v4 pipeline, ensuring fair internal comparisons.